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GEMINI: Graphene-Foil Electron-Emission Measurement Investigation for Novel Instrumentation

Amorphous carbon foils (~20-30 atomic layers in thickness) are a crucial piece of any scientific space instrument that seeks to study the solar wind or other particles present in the solar system. This is because carbon foils possess the property of releasing a secondary electron when struck by an incident particle, usually an ion. This secondary electron can be used to make coincidence or time-of-flight (TOF) measurements, which, combined with other parameters, enables scientists to deduce properties of the incident particle such as the mass, species, and energy. However, carbon foils also affect the incident particle in undesirable ways, by causing angular scattering and energy loss of the particle. It is theorized that thinner foils are able to detect lower energy particles and higher mass particles because they reduce angular scattering and energy loss, but the exact relationship between foil thickness and these other properties are not well understood. For example, previous studies have shown that the proportionality between foil thickness and angular scattering is different for carbon foils and graphene (a single atomic layer of carbon) foils, but these studies did not explore energies lower than 3 keV. In the Fall 2025 semester, the GEMINI instrument was designed to measure angular scattering and secondary electron emission of carbon and graphene foils at energies ranging from 200 eV to 2.5 keV. This Spring 2026 independent work is a continuation of the work completed in Fall 2025.

This proposal for Spring 2026 independent work proposes placing the GEMINI instrument in the Science Vacuum Chamber in the Princeton Space Physics Lab and exposing it to a variety of ion beams to gather data on the energy and yield of secondary electrons from graphene and carbon foils at ion beam energies of 200 eV to 2.5 keV. This fills a gap in existing literature exploring the angular scattering and secondary electron emission of graphene foils at low energy ion beams. The scope of work for the Spring 2026 semester includes machining the components of GEMINI, physically assembling it, and testing it in vacuum in the Space Physics Lab. For testing, GEMINI was designed to hold three carbon or graphene foils at once so that direct comparisons can be made between the different foils without having to remove the instrument from the vacuum chamber. Essentially, an identical ion beam can be used to study three different foils. For this semester, the three foils that will be tested in GEMINI are a graphene foil, 0.6 $\mu\text{g}/\text{cm}^2$ carbon foil, and 1.1 $\mu\text{g}/\text{cm}^2$ carbon foil. Each of the foils will be exposed to an ion beam of energies ranging from 200 eV to 2.5 keV in 100 eV increments. The beam compositions that will be tested are H^+ , H_2^+ , He^+ , and Ar^+ . The scope of work for Spring 2026 also encompasses developing a data collection and processing pipeline for GEMINI. This involves writing LabVIEW code to automate the process of imaging each of the three foils on GEMINI with different beam compositions and energies. MATLAB will be used to process the data from GEMINI's primary sensor, the imaging microchannel plate (MCP).

Long-term, the goal is to continue using GEMINI to contribute to heliophysicists' understanding of graphene foils by modifying the thickness and composition of graphene foils on the molecular scale and studying how different modifications affect angular scattering, energy loss, and secondary electron emission from graphene foils. This is future research that will be pursued in the 2026-2027 academic year.